APPROXIMATING A MORPHISM BY MORPHISMS OF FINITE PRESENTATION THROUGH A COFINAL SYSTEM

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Editorial note. No omissions, hints or 'immediate' steps in the counted span.

Allowed standard prerequisites (editorial).

- Cofinal/initial functors. Categories, Lemma initial-cofinal
- Characterization proposition. Limits, Proposition proposition-characterize-locally-finite-presentation

Source reference. Original proposition 01ZA: sources/stacks-chow/tags.txt, source lines 2585–2585. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original proposition 01ZC: sources/stacks-chow/tags.txt, source lines 2587–2587. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 002R: sources/stacks-chow/tags.txt, source lines 108–108. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $f : X \to S$ be a morphism of quasi-compact and quasi-separated schemes. Then there exists a direct set I and an inverse system $(f_i : X_i \to S_i)$ of morphisms schemes over I, such that the transition morphisms $X_i \to X_{i'}$ and $S_i \to S_{i'}$ are affine, such that X_i and S_i are of finite type over $\mathbf{Z}$, and such that $(X \to S) = \lim(X_i \to S_i)$.*

Proof. Write $X = \lim_{a \in A} X_a$ and $S = \lim_{b \in B} S_b$ as in Proposition 01ZA, i.e., with X_a and S_b of finite type over $\mathbf{Z}$ and with affine transition morphisms.

Fix $b \in B$. By Proposition 01ZC applied to S_b and $X = \lim X_a$ over $\mathbf{Z}$ we find there exists an $a \in A$ and a morphism $f_{a,b} : X_a \to S_b$ making the diagram

$$\begin{array}{ccc} X & \longrightarrow & S \\ \downarrow & & \downarrow \\ X_a & \longrightarrow & S_b \end{array}$$

commute. Let I be the set of triples $(a, b, f_{a,b})$ we obtain in this manner.

Let $(a, b, f_{a,b})$ and $(a', b', f_{a',b'})$ be in I. Let $b'' \leq \min(b, b')$. By Proposition 01ZC again, there exists an $a'' \geq \max(a, a')$ such that the compositions $X_{a''} \to X_a \to S_b \to S_{b''}$ and $X_{a''} \to X_{a'} \to S_{b'} \to S_{b''}$ are equal. We endow I with the preorder

$$(a, b, f_{a,b}) \geq (a', b', f_{a',b'}) \Leftrightarrow a \geq a',\ b \geq b',\ \text{and } g_{b,b'} \circ f_{a,b} = f_{a',b'} \circ h_{a,a'}$$

where $h_{a,a'} : X_a \to X_{a'}$ and $g_{b,b'} : S_b \to S_{b'}$ are the transition morphisms. The remarks above show that I is directed and that the maps $I \to A$, $(a, b, f_{a,b}) \mapsto a$

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and $I \to B$, $(a, b, f_{a,b})$ are cofinal. If for $i = (a, b, f_{a,b})$ we set $X_i = X_a$, $S_i = S_b$, and $f_i = f_{a,b}$, then we get an inverse system of morphisms over I and we have

$$\lim_{i \in I} X_i = \lim_{a \in A} X_a = X \quad \text{and} \quad \lim_{i \in I} S_i = \lim_{b \in B} S_b = S$$

by Categories, Lemma 002R (recall that limits over I are really limits over the opposite category associated to I and hence cofinal turns into initial). This finishes the proof. $\square$